Supplementary Table S7. Computational reporting, interpretability, limitations, and reproducibility

Study	Hardware	Computational information	Interpretation method	Interpretation finding	Limitations	Code	Data	Reproducibility
S01 — Wang et al. (2023)	Windows 10; NVIDIA RTX3080Ti GPU; 32 GB RAM	Python 3.9.7; PyTorch. Reported computation times for all 17 sites: LSTM = 1409.236 s; proposed model = 3370.881 s; without meteorological module = 1746.932 s; nearest-meteorological-station variant = 1809.841 s.	Ablation/sensitivity analysis of meteorological-information integration	Meteorological integration was shown to improve predictive performance; the reported R^2 gains were 0.207 versus no meteorological module and 0.129 versus use of the nearest meteorological station.	Depends on high-quality/high-frequency air-quality and meteorological data; applicability in data-scarce regions is limited; evaluated only in Beijing; increased computational/time cost; fixed 50 km edge threshold; undirected graphs do not represent directional transport.	NR	On request	Moderate
S02 — Zhou et al. (2024)	NVIDIA GeForce RTX4090 GPU	PyTorch 1.12.1; 100 training epochs; Adam optimizer; initial learning rate 0.001 with 5% decay every 5 epochs; early stopping.	NR	NR	Computational complexity may restrict province-/nation-scale deployment; the framework is designed for single-pollutant prediction and future work proposes joint multi-pollutant modelling.	Public	On request	Moderate
S03 — Huo et al. (2026)	64-core 13th Gen Intel Core i5-13490F @ 2.50 GHz; 64 GB RAM; 1 × NVIDIA GeForce RTX 4060Ti GPU	Windows 10; PyTorch 2.3.1+cu121; Python 3.8.19; batch size 32; maximum 50 iterations; Adam optimizer; learning rate 0.001; weight decay 0.005; early stopping.	NR	NR	Current graph does not model dynamically changing inter-city synergistic interactions; physicochemical reaction processes are not included.	NR	On request	Moderate
S04 — Jana et al. (2024)	Google Colab; NVIDIA Tesla P100-PCIE GPU; 16 GB RAM	Python 3.6; Keras/TensorFlow 2 (TensorFlow 2.6.0). Reported inference times/parameters: STGCN-A 30 ms (1 pollutant)/24 ms (3 pollutants), 0.88M/0.81M parameters; STGCN-B 23/22 ms, 0.68M/0.63M; STGCN-C 43/33 ms, 3.69M/3.58M.	NR	NR	NR	NR	On request	Moderate
S05 — Qi et al. (2019)	NR	NR	NR	NR	Abrupt extreme PM2.5 events, particularly values >400 $\mu g/m^3$, were not represented well; the authors indicate that additional explanatory variables are needed.	NR	NR	Cannot determine
S06 — Xiao et al. (2022)	Linux; Intel Xeon Gold 6240 CPU @ 2.60 GHz; NVIDIA RTX2080Ti GPU	All tests use a 24-time-step historical window; experiments were repeated with fixed random seeds; L2 loss used.	Graph/edge visualization and directed transport analysis	Incoming/outgoing dynamic graphs and wind-field-driven directional connections were analysed to demonstrate time-varying pollutant transport.	NR	NR	Public	Moderate
S07 — Sui and Han (2023)	Linux server; Intel Xeon Platinum 8260 CPU @ 2.40 GHz; NVIDIA TESLA V100 16 GB GPU	History window = 12 h; forecast window = 6 h; hidden size = 64; batch size = 256; up to 500 epochs with early stopping.	Ablation analysis	Removing either spatial view, the dual-view attention fusion, or the auxiliary task degraded performance, supporting the contribution of each component.	Future work is proposed to consider more environmental factors and improve prediction across broader settings; no detailed standalone limitations section was reported.	NR	On request	Moderate
S08 — Teng et al. (2023)	NR	NR	Graph/edge contribution analysis and source-receptor visualization	Time-varying edge weights were used to quantify changing contributions from surrounding cities to target-city PM2.5 forecasts and to interpret regional pollutant transport.	Spatial coverage focused on BTH and surrounding cities and could be expanded farther for long-range transport; the study focused on PM2.5 and proposes simultaneous prediction of additional pollutants.	NR	Public / links shared in the manuscript	Moderate
S09 — Zhou et al. (2021)	NR	TensorFlow is reported; other hardware/training-time details = NR	Physics/domain-based graph interpretation using dynamic wind-field edges	The dynamic directed graph visualizes likely pollutant-transport relationships and provides spatial interpretability linked to wind fields.	Validation should cover more regions and time periods; graph construction uses wind as the principal factor and should include multiple factors; temporal trend/periodicity are not fully modelled; physical mechanisms should be integrated more deeply.	NR	NR	Cannot determine
S10 — Zhou et al. (2022)	NR	NR	GNNExplainer; Granger-causality analysis	Model-selected influential station relationships were compared with Granger-causal and wind-connected relationships to interpret spatial source-receptor dependencies.	NR	NR	NR	Cannot determine
S11 — Chen et al. (2025)	NVIDIA 3090Ti GPU	Python; PyTorch; batch size 64. Other timing/model-size information = NR.	Graph/edge contribution and directional transport visualization	The model analyses changing regional influence among cities and links forecast behaviour to dynamic transport pathways.	Monitoring-data dependence limits application in unmonitored areas; graph criteria/thresholds are fixed; abrupt complex pollution events and unmodelled variables remain challenging; broader regional validation and more robust data alignment are needed.	NR	Public / data-source links shared in article	Moderate
S12 — Guyu et al. (2025)	NR	NR	Ablation analysis	Ablation experiments quantified the contribution of the mixed graph-convolutional GRU and self-attention components, with the self-attention component particularly improving longer-horizon prediction.	The static distance-based graph cannot fully represent real-time environmental changes; prediction accuracy for high pollution grades is relatively low; future work proposes dynamic graph construction.	NR	Restricted / confidential	Low
S13 — Shi et al. (2026)	NVIDIA RTX3090 GPU	Ubuntu 20.04; Python 3.8; PyTorch 2.0.0. Reported AST-GT time per epoch: Beijing = 11.3716 s; India = 12.5023 s.	NR	NR	The data-driven framework does not explicitly impose physical constraints for diffusion, transport, and deposition; broader climate/station validation and cross-region transfer are future needs; extension to additional pollutants is proposed.	NR	Public	Moderate
S14 — Zhao et al. (2026)	NR	Hyperparameters for all models were optimized using Bayesian optimization for 50 iterations on the validation set; other hardware/training-time details = NR.	Fuzzy Tsallis Entropy station-importance analysis; spatial importance mapping	The FTE branch identifies influential/key monitoring stations; high-importance stations were reported to cluster in urban core areas with dense human activity such as transport, commercial, and residential zones.	Current spatial network is static and geographically based and does not update with dynamic meteorology; this limits sudden/local-event modelling and may increase overfitting risk; only 2022 data were used, so interannual generalization is untested; integration of fuzzy entropy requires further optimization.	NR	Restricted / confidential	Low
S15 — Ni et al. (2023)	NVIDIA GeForce RTX3090 GPU; Intel Xeon Silver 4210 CPU	Ubuntu 20.04; CUDA 11.6. ASGCN reported 401,166 parameters; single-step training time = 721.56 s (KnowAir) and 379.37 s (UrbanAir).	Ablation analysis	Ablation experiments show positive contributions from the adaptive/static graph components and temporal module, with contribution size varying by dataset.	Physical/aerodynamic mechanisms, natural factors, air diffusion, and geographic information are not fully incorporated; time complexity should be further reduced; interpretability of the local dynamic graph should be improved.	NR	On request	Moderate
S16 — Kim et al. (2023)	NR	NR	Ablation-based analysis	Ablation results showed that PM2.5-similarity edges and graph-attention components improve prediction, including reported improvement in detection performance.	Prediction is less accurate during high-pollution events because the underlying relationships become more complex; future work proposes extension to other pollutants.	NR	On request	Moderate

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S7 continued

Study	Hardware	Computational information	Interpretation method	Interpretation finding	Limitations	Code	Data	Reproducibility
S17 — Zeng et al. (2025)	Windows 10; NVIDIA RTX3080Ti GPU	Visual Studio Code; history window = 24 steps; learning rate = 1e-3; batch size = 32; MSE loss; early stopping. Exact training/inference times are shown graphically but not reported numerically in the main text.	Learned-adjacency visualization; ablation analysis	The trained adjacency matrix is visualized; strong self-loops indicate the importance of local history, and ablations show the contribution of adaptive graph generation and jump graph convolution.	Long-term performance declines; performance varies substantially by pollutant; prediction differs by season, especially summer versus winter; only single-task/single-pollutant prediction is used in the main model.	NR	Public (Beijing dataset explicitly described as public; Chongqing data from CNEMC)	Moderate
S18 — Liu et al. (2026)	NR	Per-step complexity = O(MdH + Mhs + d^2 + K(Ld + L^2) + h_g(d + Kd_e)); dominant memory = O(KL^2). Reported model size = 82×10^3 parameters; initial offline training = 1.4 s/epoch; online update = 0.42 ms/step; inference = 0.36 ms/step; peak memory = 0.24 GB.	Adaptive source-weight analysis; latent interaction-matrix visualization	Adaptive source weights increase the contribution of pollutant information during severe episodes, and the learned soft interaction matrix reveals time-varying predictive coupling among latent components.	Needs validation across more cities/countries/sensor settings; current drift analysis uses operational segmentation rather than controlled drift benchmarks; forecasts are deterministic point predictions without probabilistic uncertainty; richer feature attribution is needed; study focuses on PM2.5 rather than multi-pollutant forecasting.	NR	On request	Moderate
S19 — Xu et al. (2023)	Linux; Intel Xeon Gold 5218 CPU @ 2.30 GHz; GPU reported as 'NVIDIA Corporation Device 2204 (rev a1)'	Maximum 50 epochs; early stopping patience = 10; RMSProp; learning rate = 5e-4; weight decay = 5e-4; batch size = 32; experiments repeated 10 times.	Adaptive edge-attribute visualization; complex-network analysis	The study compares physical wind edges and learned adaptive edges and analyses incoming/outgoing edge weights and node centrality; learned adaptive edges reveal hidden spatial relationships not captured by the physical graph.	Future meteorology is represented using ERA5 reanalysis and may mismatch real operational forecasts; learned edge parameters are a black box; monitoring stations are unevenly distributed and some highly polluted areas are unsampled; performance is weak for extreme heavy-pollution events.	NR	Public	Moderate
S20 — Ma et al. (2026)	NR	Input length = 72 h; validation-set early stopping; other hardware/training-time/model-size details NR	Learned graph-level weight analysis / visualization	The relative importance assigned to distance, neighborhood, distribution-similarity and functional-similarity graphs changes by dataset and forecast horizon, showing non-uniform contributions from the four graph views.	The strongest advantage occurs in high-variance/high-pollution regimes and the model does not always dominate low-variance regimes; DMGENet requires reasonably complete input records across all four graphs; systematic data gaps increase residuals; RLMC is an approximate graph-level integration strategy rather than a theoretically complete sequential decision model.	Public	Public	High
S21 — Su et al. (2023)	NR	NR	Ablation/sensitivity analysis of meteorological data, distance graph, POI graph and temporal attention	Removing meteorological features or either graph view degraded performance; the study uses these ablations to demonstrate the contribution of geographic, meteorological and POI context.	NR	NR	On request	Moderate
S22 — Zhao et al. (2025)	NR	Learning rate=0.001; batch size=64; hidden dimension=64; two layers; other hardware/training-time information NR	NR	NR	Runtime is approximately twice that of GAT-GRU; the study performs single-pollutant forecasting and the authors propose adding other pollutants/environmental factors; the clustering hypergraph may not fully reflect regional differences; directional influence/transmission pathways are not clearly represented.	NR	On request	Moderate
S23 — Gao et al. (2026)	Intel(R) Core(TM) i9-13900HX CPU @ 2.20 GHz; 16 GB memory; NVIDIA A100-PCIE GPUs; Windows 10	Anaconda; Python 3.9; PyTorch; previous 6 time steps used to predict the next step; learning rate=1×10⁻5; weight decay=1×10⁻3; batch size=4; Adam optimizer; 30 training epochs	Learned hypergraph/graph relationship analysis; ablation analysis	The learned hyperedge and pairwise relations provide soft station-association structure; station-level R^2 visualization and ablations are used to assess the benefit of adaptive higher-order spatial modelling.	Abrupt non-periodic fluctuations remain difficult in mountainous/plateau regions; topographic features are not explicitly incorporated; $O(N^2)$ complexity from dynamic graph construction and attention recalculation limits scalability in large networks.	NR	Public	Moderate
S24 — Zhang et al. (2024)	NR	RMSprop; learning rate=0.0003; weight decay=0.003; batch size=32; maximum 50 epochs; early stopping=10; MSE loss. Hardware/training-time information NR.	NR	NR	NR	NR	NR	Cannot determine
S25 — Mandal and Thakur (2023)	Google Colab Pro; GPU model NR	PyTorch; RMSprop; learning rate=0.0015; weight decay=0.0001; 50 epochs; L1 regularization. Edge-MLP parameters=1566; node-MLP parameters=56; cluster-wise GRU hidden dimension=25; temporal GRU hidden dimension=32.	Ablation analysis; station-wise/spatial error and forecast-map analysis	Ablations assess the roles of clustering, spatial attention and temporal modelling; the paper also visualizes spatial PM2.5 forecasts and station-level behaviour.	Increasing historical input information increases computational expense; ERA5 is not free of charge for operational real-time use; the authors note the coarser temporal resolution of GFS as an operational alternative and propose further development including dynamic graph clustering.	NR	On request	Moderate
S26 — Hu et al. (2024)	Single NVIDIA RTX 3090 GPU	Python; PyTorch 1.13.1; historical input length=24; prediction length=24; learning rate=0.0005; Adam optimizer. AHGCNN parameters=208,862; training time=46 s/epoch; reported average MAE=15.84 in the computation-cost table.	Ablation analysis; learned-adjacency/hierarchical-component analysis	Ablations show that the adaptive graph structure and hierarchical attention aggregation improve forecasting relative to variants without these components.	NR	NR	On request	Moderate
S27 — Tan et al. (2022)	NR	Reported computation time for the proposed model = 201.471 s; other hardware/software information NR	NR	NR	NR	NR	On request	Moderate
S28 — Gao and Li (2021)	NR	Input history length=4 h; maximum epochs=1000; learning rate=0.05; weight decay=0.0005; early-stopping patience=100; MAE loss. First GLSTM layer parameters=2704; first fully connected layer parameters=15. Software/framework and hardware NR.	Learned adjacency-matrix visualization	The learned parameterized adjacency is visualized to show the relative importance of surrounding stations; the matrix is asymmetric, indicating directional station-to-station influence learned from the prediction task.	The paper notes that matching each air-quality station only to its nearest meteorological station does not fully represent a heterogeneous monitoring system and proposes heterogeneous graph neural-network modelling as future work.	NR	NR	Cannot determine
S29 — Liu et al. (2021)	NR	Python 3.6.5; Keras 2.2.0. Q-learning maximum iteration=50; learning rate=0.95; discount parameter=0.5; optimization objective=RMSE.	Graph/correlation analysis	The paper analyses the Pearson-correlation structure among six nodes and reports strong correlations among Nodes 1–3 and additional correlations linking Nodes 4–6 to the core sites.	The authors state that air-quality data change dynamically in practical applications and the model needs real-time updating; practical deployment would require embedding the model in a relevant system to maintain timeliness.	NR	NR	Cannot determine
S30 — Zhang et al. (2024)	Intel(R) Xeon(R) Silver 4210 CPU @ 2.20 GHz; GeForce RTX 3090 GPU; Ubuntu 16.04	Initial learning rate=0.01; training rounds=120; model implemented for dense 456-node hourly data. Software framework name NR in the extracted experimental setup.	Sensitivity/ablation analysis of graph threshold and monitoring-station density	β=0.8 gave the best combined prediction efficiency; increasing monitoring-station density improved fine-grained urban PM2.5 inference.	The authors state that additional influential factors such as urban structure, meteorological conditions and traffic environment should be incorporated in future work to further improve prediction.	NR	NR	Cannot determine

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S7 continued

Study	Hardware	Computational information	Interpretation method	Interpretation finding	Limitations	Code	Data	Reproducibility
S31 — Ma et al. (2022)	NVIDIA Quadro P6000 GPU (24 GB); Intel Xeon Gold 5118 CPU; 128 GB RAM	PyTorch; graph kernel=3; temporal kernel=3; dropout=0.3; learning rate=0.001; batch size=32; epochs=350; Adam optimizer.	Input/fusion-combination analysis	The paper compares different heterogeneous-data combinations and reports the best performance when all four observation sources are fused.	The method requires explicit monitoring-point locations; the fully connected graph becomes computationally expensive as node count grows; only four observations/data types were evaluated, making the effects of combinations difficult to interpret comprehensively.	NR	Public	Moderate
S32 — Gu et al. (2025)	NVIDIA RTX 4090D 24 GB GPU; Ubuntu 20.04.3	PyTorch and PyTorch Geometric; vectorized/vmap parallel processing. Original and transfer models have >2.3 million total parameters, but transfer fine-tuning trains about 1/500 of the parameters; training data requirement ratio=1× transfer vs 5× original; training-time ratio=1× transfer vs ~50× original.	Dynamic edge-weight visualization and wind-pattern/transport interpretation	The paper visualizes changing graph weights alongside wind conditions and interprets high/low edge weights as time-varying pollutant-transport influence among stations.	NR	NR	On request	Moderate
S33 — Zhang et al. (2024)	Intel Xeon Silver 4210 CPU @ 2.20 GHz; 256 GB RAM; 4× NVIDIA RTX 3090 GPUs; Ubuntu 20.04.5	Python 3.7.15; PyTorch; 1,997,280 records; history length=60 h; epochs=100; loss=MAE; attention heads=8; Adam optimizer.	Ablation analysis of multi-graph components and spatiotemporal modules	Ablations show that temporal information is essential and that adding functional and adjacency subgraphs improves the multi-graph model over pruned variants.	The authors state that hidden geographical/topographical conditions are not fully considered and that regional spatial heterogeneity may affect graph relationships; future work proposes adding more spatial/contextual data and improved functional-region graph construction.	Public	Public	High
S34 — Zhao et al. (2025)	NR	NR	Dynamic graph/geographical-matrix analysis and ablation experiments	Ablations show that distance, learned geographical relationships and dynamic wind information each contribute to performance; the authors interpret the learned static matrix as inherent geographical influence and the wind graph as time-varying transport.	Increasing the number of stations raises GNN computational complexity; dynamic graphs hinder computational parallelism; generalizability beyond East China is untested; the authors propose adding detailed topography/urban features, improving efficiency, and better integrating spatial and temporal modules.	NR	Unclear from full text — the methods explicitly use CNEMC and ERA5 data, but the formal Data availability statement says "No data was used for the research described in the article."	Cannot determine
S35 — Zhang et al. (2025)	NVIDIA RTX 3090 GPU	Python 3.8.13; PyTorch. Reported training time per epoch for the proposed model: 24 h=753.08 s; 48 h=910.06 s; 72 h=1203.58 s.	Ablation analysis of spatial graph, spatiotemporal graph, support embedding and SCIBlock	Ablations show that the learned spatiotemporal graph, support embedding and multi-granularity SCIBlock each contribute to performance; a simple static graph alone can be inferior to removing spatial information at longer horizons.	The authors acknowledge relatively high computational cost and identify reduction of computational overhead as future work; they also propose including additional ground-level data.	Public	Public	High
S36 — Ye et al. (2025)	NVIDIA RTX 3060 GPU; Intel Core i5-11400 CPU @ 2.60 GHz	Python 3.10.0; PyTorch 2.0.0; training epochs=20; experiments repeated five times; early-stopping patience=5; batch size=32.	Instantaneous phase synchronization relevance analysis	IPS identifies surrounding cities whose PM2.5 temporal variation is synchronized with the target; the study shows that spatial relevance is not determined solely by geographical proximity.	The model relies heavily on wind speed/direction, latitude, longitude and altitude; it does not include temperature, humidity, precipitation, NOx, SO2 or CO, nor land use/vegetation or temporal factors such as seasonal variation/time of day. The authors also state that the IPS estimator is tailored to the Transformer framework and may have limited direct applicability to other baselines.	Public	Public	High
S37 — Velez-Serrano and Alvaro-Meca (2026)	Linux desktop; NVIDIA RTX 3090 GPU; Intel i9-1490 CPU (model string preserved as reported); 192 GB RAM	Python. ISTGT maximum training=600 epochs; early-stopping patience=100; batch size=4; Adam optimizer; cosine-annealing schedule; gradient clipping. Reported ISTGT time≈80 s/epoch; convergence approximately one full day.	Graph-weight interpretation; temporal attention visualization	The adaptive graph explicitly incorporates distance, altitude and population, while attention visualizations are used to examine temporal contributions to PM2.5 prediction.	Performance is weaker for anomalous/extreme events such as African dust intrusions; operational use depends on periodically updated public data and model retraining; training/convergence can take approximately one day. The authors propose incorporating real-time meteorology, air-quality APIs and satellite information.	Public	Public	High
S38 — Dong et al. (2024)	NR	Code is provided; Adam optimizer; maximum 200 epochs; MAE training objective with additional KL regularization/variational inference; hyperparameters tuned on validation set. Hardware NR.	Ablation analysis; token-mixer/SIM component analysis	Experiments show that the spatial information modulator captures global spatial information; the authors report that simple pooling can replace attention while reducing learnable parameters and maintaining forecasting effectiveness.	The authors explicitly state that, due to experimental-condition limitations, the research was carried out only in a small-scale environment and future work should apply the method to real-life scenarios.	Public	On request	Moderate
S39 — Dun et al. (2022)	NR	SGD optimizer; learning rate=0.01; momentum=0.9; weight decay=5×10^-4; dropout=0.5; batch size=256; 200 epochs.	NR	NR	The authors state that rainfall, vehicle exhaust emissions, and human activities are not fully considered and propose further development of dynamic adjacency/attention mechanisms.	NR	On request	Moderate
S40 — Tariq et al. (2023)	NR	MATLAB used for preprocessing; PyTorch Ray Tune/random search for hyperparameter selection; 100 epochs; Adam optimizer; MSE loss; two hidden layers with 64 GRU units; dropout=0.25; embedding dimension=10; learning rate=0.003.	NR	NR	Larger distances between stations increase prediction discrepancy; neighbouring nodes can operate at different monitoring time scales, causing low-quality inputs.	NR	On request	Moderate
S41 — Wang et al. (2024)	NR	Python; early stopping applied. Detailed model parameters are reported in supplementary material not included in the supplied main PDF.	Attention-matrix visualization; graph/community interpretation	Cold- and warm-season GAT attention matrices quantify inter-city pollution influence; normalized attention weights are used to infer regional connections and support season-specific collaborative-control groupings.	Prediction accuracy/reliability decreases with forecast time, particularly for long-term PM2.5. The fixed 200-km physical threshold is a simplification; the authors propose more dynamic adjacency construction using hourly meteorology/geography/pollution and city-specific improvement.	NR	On request	Moderate
S42 — Zhang et al. (2024)	NR	Python; PyTorch; RMSprop optimizer; 50 epochs; early stopping; batch size=32; GWO population=10 wolves and 10 iterations; optimized parameters include attention heads, hidden units, learning rate and regularization.	Ablation analysis	Ablation results quantify the contribution of multi-GAT spatial extraction and GWO hyperparameter optimization.	NR	NR	On request	Moderate

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S7 continued

Study	Hardware	Computational information	Interpretation method	Interpretation finding	Limitations	Code	Data	Reproducibility
S43 — Liu et al. (2024)	NR	NR	NR	NR	The benchmark covers a relatively short September–December period, which accentuates distribution shift; graph construction based on historical geographical/correlation relationships can become imprecise or outdated when the data distribution changes.	Public	Public	High
S44 — Mohammadzadeh et al. (2024)	NR	NR	NR	NR	The study is limited to six Michigan locations; authors propose testing the model in other regions with different climatic conditions and industrial landscapes and integrating additional environmental/socio-economic factors and real-time data.	NR	On request	Moderate
S45 — Liao et al. (2024)	NR	NR	Scenario/counterfactual fire simulation	The model is used to compare PM2.5 predictions with and without prescribed-fire/wildfire contributions and to estimate how prescribed-fire timing can alter future wildfire-related PM2.5 exposure.	The forecast models can tend to output values close to the previous observation; the authors propose incorporating additional physics-based modelling into the GNN framework in future work.	Public	Public	High
S46 — Han et al. (2025)	NR	NR	NR	NR	The paper explicitly lists limited generalization beyond the studied datasets; data quality/availability; incomplete representation of complex atmospheric processes and spatial/temporal dynamics; computational demands; black-box interpretability; external factors such as policy/economic activity/wildfires/industrial accidents; overfitting/algorithmic bias; lack of dynamic feature selection; and limitations of standard evaluation metrics.	NR	On request	Moderate
S47 — Huang et al. (2025)	NR	Huber loss is used; additional hardware/framework details NR in the supplied PDF.	NR	NR	Due to computational-resource limitations, only major cities were selected for modelling. The authors propose incorporating seasonal variation, industrial emissions, vehicle emissions and other factors to improve future forecasts.	NR	NR	Cannot determine
S48 — Wang et al. (2024)	NR	PyTorch; 24-h historical input window; Adam optimizer (printed as "Adma"); MSE loss; initial learning rate=0.01 with dynamic adjustment; 100 epochs; batch size=256; hyperparameters selected by grid search.	NR	NR	The paper notes limited quantity and uneven spatial distribution of stations, poorer stability of CNN-LSTM under this setting, reduced effectiveness at pollutant-concentration peaks, and decreasing accuracy for longer prediction sequences; future work should improve long-sequence prediction.	NR	Public	Moderate
S49 — Dimri et al. (2024)	NR	Batch size=64; 100 training epochs; weight decay=0.0005; early stopping threshold=10 epochs; learning rate=0.001. Past 24 h + future 72 h form a 96-h spatiotemporal input span.	Learned adjacency/attention analysis	The authors interpret adaptive node connectivity and attention scores as data-driven station influence; AGATNet learns connections without predefined distance, geographic barriers, or advection coefficients.	AGATNet occasionally underestimated PM2.5 spikes during forecast days 2 and 3 because there is no defined mechanism restricting connections, which can cause GAT to misallocate attention scores to critical versus less-important stations.	Public	Public	High
S50 — Qi et al. (2024)	NR	Five-fold cross-validation is used for hyperparameter selection; exact training/inference time and model size NR.	Ablation analysis	Ablations show that both the GAT spatial component and the multi-period temporal component materially contribute to prediction performance.	The authors state that future design should focus on improving model generalization and reducing training time.	NR	NR	Low
S51 — Pan et al. (2024)	NR	The model has more than 20,000 parameters; experiments were replicated five times. The paper reports average training-time/model-efficiency discussion but specific hardware is NR.	Permutation feature-importance analysis; graph-message-passing ablation	Permutation tests identified 2 m temperature and boundary-layer height as the most influential meteorological variables. Shuffling 2 m temperature increased RMSE by 3.28 and MAE by 2.95; shuffling boundary-layer height reduced CSI by 6.99% and POD by 5.66%.	The model is less efficient than simpler competitors because of its more complex structure; authors identify model efficiency as an area needing improvement.	NR	On request	Moderate
S52 — Jin et al. (2023)	12th Gen Intel Core i5-12500H processor, 4800 MHz; 16 GB memory	PyTorch; Bayesian-optimized hyperparameters: GraphSAGE layers=4; BGraphGRU layers=1; epochs=11; learning rate=0.0327; edges_per_node=3.	NR	NR	NR	NR	Public	Moderate
S53 — Zhang et al. (2022)	NVIDIA Quadro P6000 GPU with 24 GB memory	PyTorch tools used for machine-learning models; TensorFlow configured for deep-learning models; Adam; learning rate=1e-4; batch size=32; max epochs=200; MSE loss; lookup/window size=24.	NR	NR	The method is designed for a single monitoring station in a single city and therefore fails to analyze spatial variations; the authors propose extension to regional multi-station/multi-city PM2.5 estimation.	NR	On request	Moderate
S54 — Zeng et al. (2022)	NR	PyTorch; Adam; learning rate=0.001; batch size=128; 100 epochs; 3 STGODE blocks per module and 2 STGODE modules. Average training time=63 s versus 76 s for the strongest comparison model.	Ablation analysis	Ablation results show that the wind-informed adjacency improves over distance-only construction and that the external air-humidity fusion module further improves PM2.5 prediction.	The dataset has only 9 monitoring stations and relatively simple spatial relationships, limiting the magnitude of performance gains; the authors plan validation on datasets with more stations.	NR	On request	Moderate
S55 — Lu et al. (2025)	NVIDIA RTX 3090 GPU	PyTorch; hyperparameters selected by grid search on validation set; all results averaged over five independent random seeds; paired t-tests used for significance.	Learned station-group visualization; ablation analysis	The learned soft assignment groups stations by functional similarity rather than geographic proximity alone; the selected three groups exhibit distinct variance/error characteristics, and removing the adaptive adjacency increased 24-h RMSE by approximately 23.82%.	The current model uses fixed hourly, daily, and weekly temporal scales and requires further computational-efficiency optimization; authors plan extension to other urban areas and adaptive period identification such as FFT.	NR	Public	Moderate
S56 — Abbasi et al. (2025)	Windows 11; NVIDIA GeForce RTX 3050 Ti GPU; 16 GB RAM	Python 3.10.4; PyTorch 2.4.1; PyTorch Geometric 2.6.1. ClusLite-STGCN-GRU complexity: FBP=337.27 s; FP=3.66 ms; epoch time=373.41 s; memory=720.09 MB; FLOPs=16,549,904; parameters=257,288.	Copula dependency analysis; pollutant-specific station clustering	Kendall's τ with PM2.5: PM10=0.665; NO2=0.424; SO2=0.388; CO=0.386; O3=-0.262. Spatial clustering produced five PM2.5 clusters, four PM10 clusters, and five NO2 clusters; O3/CO/SO2 each formed one cluster and were excluded from final spatial inputs.	Direct comparison with earlier Tehran studies is limited by differences in study periods, features, data quality, and model configurations; the authors also identify broader geographic validation/generalizability as future work.	NR	Public	Moderate

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S7 continued

Study	Hardware	Computational information	Interpretation method	Interpretation finding	Limitations	Code	Data	Reproducibility
S57 — Guan et al. (2025)	Windows 11; Intel i7-13700 CPU (5.0 GHz); NVIDIA GeForce RTX 4060 GPU with 8 GB; 32 GB RAM	Python 3.9; Pandas 2.2.3; NumPy 2.2.6; PyTorch 2.7.0; maximum training epochs=300; early stopping if validation loss fails to improve by at least 1×10^-4 over 50 epochs; STGCA reported time=1602.0 s versus ASTGCN=1855.2 s, MTGNN=2160.4 s, MTGODE=2450.9 s.	Ablation analysis of graph filter/convolution and attention mechanism	Ablations demonstrate gains from the proposed DRWF/improved convolution and spatiotemporal attention; the study also reports that STGCA maintains more stable accuracy as forecast step increases.	The model has difficulty distinguishing precise boundaries of sharp concentration gradients and capturing extreme values. Validation was conducted only in Beijing; authors also identify unperformed feature-importance analysis and uncertainty quantification as future improvements.	Public	Public	High
S58 — Chen et al. (2024)	NR	Bayesian automatic hyperparameter search; Adam optimization. Detailed hardware/training time NR.	NR	NR	The model is relatively complex with many parameters; Bayesian hyperparameter tuning is time-consuming/serial; the POI information is treated as constant although real urban conditions may change.	On request	Public	Moderate
S59 — Chauhan et al. (2026)	Single NVIDIA GPU; model NR	Python/PyTorch; Adam; learning rate=0.0005; weight decay=1e-5; batch size=32; maximum 150 epochs; hidden dimension=128; 4 GAT heads; dropout=0.2; early stopping patience=20; best epoch=83 and stopping epoch=103; complexity discussion includes O(N^2+TNF).	Feature attention	The feature-attention mechanism highlights environmental-variable importance; PM10, NO2 and temperature are reported among influential variables. The authors caution that attention-based feature importance is not causal attribution.	Fixed geographical graph; unknown generalization beyond the evaluated network; offline training/inference; quadratic GAT scaling; sparse variants were not benchmarked; feature attribution is not causal; statistical significance/confidence intervals were not reported.	NR	Public	Moderate
S60 — Li et al. (2023)	Ubuntu server; Intel Xeon CPU 2.4 GHz; 128 GB RAM; NVIDIA RTX 3090 GPU with 24 GB GPU RAM	Three fully connected dimensions reported as 9, 100, 1; Chebyshev GCN K=2; Informer has 2 encoder layers, 1 decoder layer, model dimension=512, feed-forward dimension=2048, 8 heads; learning rate=0.0001; batch size=32; MSE loss; Adam; 30 epochs; early-stopping patience=5.	Ablation analysis	Component comparisons are used to evaluate the contribution of MLP/GCN spatial processing and Informer temporal modelling.	NR	NR	On request	Moderate
S61 — Mirzaei and Torghabeh (2026)	NVIDIA RTX 3090 GPU (24 GB)	Average training time≈8.5 s/epoch; GPU allocated memory≈0.22 GB; Euler integration≈2–3 s/epoch; RK4≈8–9 s/epoch. Fully connected spatial message passing has O(N^2) scaling.	Graph/edge interpretation through physically meaningful edge attributes	Spatial relations are represented through distance, bearing and wind-dependent features so learned station interactions can be related to atmospheric transport rather than a purely abstract adjacency.	Only five Beijing stations; fully connected graph produces O(N^2) scaling; pollutants are modelled as correlated signals without explicit chemical interactions; RK4 is 3–4× slower than LSTM-style training; scaling increases memory/inference requirements; wind fields are simplified; future wind is assumed available without modelling forecast uncertainty.	On request	Public	Moderate
S62 — BADOUCH and BELHOUCINE (2026)	Apple M1 used for preliminary compression/inference experiment; main training hardware NR	Adam; learning rate=1e-3; weight decay=1e-5; MSE loss; batch size=64; 24-h input history; early-stopping patience=7; seeds 42/123/777. Reported latency: GCN+Transformer=1.11 ms/batch; Linear+Transformer=1.03 ms/batch; GCN+LSTM=4.74 ms/batch. Preliminary 8-bit quantization on Apple M1 reduced inference latency by 42%.	NR	NR	Single real dataset; negative GCN finding is Beijing-specific and not yet cross-city replicated; aggregation can favor ARIMA; uncertainty quantification is limited to exploratory three-seed intervals; Apple M1 is not representative of edge hardware and quantization accuracy impact was not fully validated.	Unclear from full text — described as full/public code availability, but repository URL is withheld during review	Public	Moderate
S63 — Huang et al. (2025)	NR	On Dataset 3, reported runtime=1001.20 s; FLOPs=52.008 G; parameters=0.090 M.	Ablation analysis	Ablations assess the contribution of global dynamic dependency learning, local physical/transport graph modelling and their fusion.	NR	NR	Public	Moderate
S64 — Pei et al. (2022)	Windows 10; Intel Core i5-10500H CPU at 2.5 GHz; no GPU reported	Python; RMSProp; MSE loss; batch size=24; epochs=150; history/window=24 h. Hyperparameter table reports initial learning rate=0.001, whereas the experiment text states 0.0007 with a 0.5 decay every 40 epochs — Unclear from full text.	Ablation analysis	Ablations quantify the contribution of multivariate graph learning, GRU temporal modelling and AVMD decomposition.	NR	NR	Public	Moderate
S65 — Yu et al. (2026)	NR	PyTorch; Adam; learning rate=0.001; batch size=65; 30 epochs; early stopping; latent phase dimension=32; 24-h historical input and 12-h output.	Attention visualization; graph/edge interpretation; Koopman spectral/eigenfunction interpretation	Learned attention reveals hub-like stations, regional clusters and varying neighbor/self influence within 200 km; Koopman modes/eigenfunctions are interpreted as latent growth/decay patterns and characteristic time scales.	NR	Public	Public	High
S66 — Wang et al. (2024)	NR	Adam; learning rate=1e-4; batch size=512; maximum 100 epochs; MSE loss.	NR	NR	NR	NR	On request	Moderate
S67 — Zhao (2025)	NR	SHAP analysis implemented in Python 3.7; comparison-model hyperparameters tuned by grid search. Other computational details NR.	SHAP	PM10 and CO are reported as major positive contributors to PM2.5; NO2 also contributes positively; temperature has a significant negative effect, while humidity shows an indirect positive contribution.	NR	NR	On request	Moderate
S68 — Zeng et al. (2021)	NR	PyTorch; learning rate=0.001; Adam optimization; MSE loss; Train:Validation:Test=8:1:1.	NR	NR	The authors state that PM2.5 is also affected by additional factors such as weather conditions and wind speed that were not fully included and should be considered in future work.	NR	Public — the paper states that data supporting the findings are included within the article	Moderate
S69 — Wang et al. (2022)	Single NVIDIA 2080 Ti GPU with 11 GB memory	PyTorch; 100 epochs; learning rate=0.0005; batch size=64.	Ablation analysis of meteorological knowledge	Removing meteorological node attributes substantially worsened RMSE/MAE/CSI, supporting the contribution of meteorological domain knowledge; the paper does not separately analyze the effect of each meteorological variable.	Different meteorological variables are used jointly and their individual effects on PM2.5 prediction are not specifically analyzed; the authors identify this as future work.	NR	Public — the paper explicitly describes KnowAir as a public dataset	Moderate

Note: Hardware, computational information, interpretability methods and findings, author-reported limitations, and access/reproducibility fields reproduce the supplied extraction. Interpretability techniques may include ablation; their listing does not independently validate explanations. Reproducibility ratings are qualitative labels in the extraction, not standardized risk-of-bias scores. NR denotes not reported.